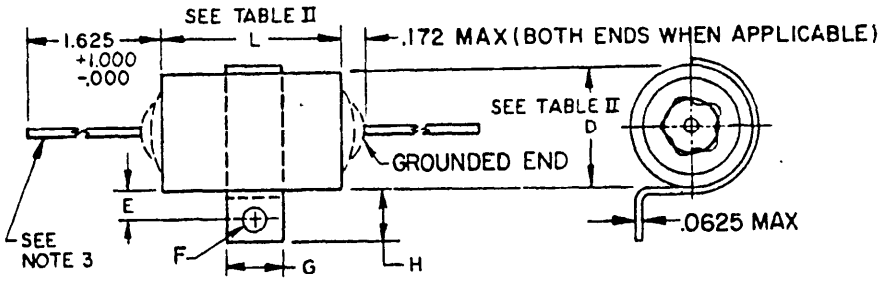
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, PLASTIC (OR PAPER-PLASTIC) DIELECTRIC, AXIAL-WIRE TERMINAL, TANGENTIAL RETAINER, TUBULAR (UNINSULATED), (HERMETICALLY SEALED IN METAL CASES), STYLE CQ10

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-19978.

INACTIVE FOR NEW DESIGN AFTER
30 August 1974



INCHES	MM	INCHES	MM	INCHES	MM
.001	.03	.0625	1.59	.312	7.92
.004	.10	.144	3.66	.400	10.16
.005	.13	.156	3.96	.438	11.13
.025	.64	.172	4.37	.500	12.70
.031	.79	.188	4.78	.562	14.27
.032	.81	.235	5.97	1.000	25.40
.062	1.57	.250	6.35	1.625	41.28

RETAINER DIMENSION

Case dimension D	H±.062	G±.062	E±.031	F±.005
Less than .562	.312	.250	.188	.144
.562 and up	.438	.500	.250	.156

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Leads shall be of tinned solid wire, .025(No. 22 AWG) for cases .235 and .312 in diameter; and .032(No. 20 AWG) for cases .400 diameter and above. Tolerance on all lead wire diameters shall be +.004, -.001.

FIGURE 1. Style CQ10 capacitors.

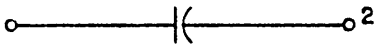
FSC 5910

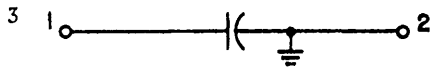
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REQUIREMENTS:

Design and construction:
Dimensions and configuration - See figure 1 and table II.
Case material - Nonmagnetic (end seal may be of magnetic material).
Dielectric material - Paper and polyethylene terephthalate (characteristic K).
Terminals - Axial-wire lead (see figure 1).

Circuit diagram - 1





Rated voltage - See table II.
Rated temperature - -65° to +125°C.
Capacitance (Cap.) (nom) - See table II.
Capacitance tolerance - ±10 percent (symbol K).
Vibration grade - 3.
Dissipation factor (DF) (max) - 1.0 percent.

Seal: Method 112 of MIL-STD-202, test condition letter - A.

Barometric pressure: In accordance with MIL-C-19978.
Test points:
Circuit diagram 1 - Between terminals and case.
Circuit diagram 3 - Between ungrounded terminal and case.

Insulation resistance (IR):
Terminal to terminal - See table I.

TABLE I. Terminal-to-terminal insulation resistance.

Capacitance rating	Minimum insulation resistance
	At 25°C
0 to 0.6 microfarad - - - - - - - - -	25,000 megohms
Greater than 0.6 microfarad - - - - -	15,000 megohm-microfarads 1/
	At 125°C
0 to 0.08 microfarad - - - - - - - - -	250 megohms
Greater than 0.08 microfarad - - - -	20 megohm-microfarads 1/

1/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

Terminals to case - Greater than 10,000 megohms.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B, with the following exception:
Direction and duration of motion - 4 hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cylindrical axis.

Salt spray: In accordance with MIL-C-19978.

Solderability: In accordance with MIL-C-19978.

Terminal strength: Method 211 of MIL-STD-202, test condition letter - D.

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TABLE II. Style CQ10 capacitors.

Part number 1/	DC voltage rating	Capacitance	Case dimensions (in inches)		
			D +.010 -.020	L ±.015	
				Circuit 1	Circuit 2
	Volts	µF			
CQ10A-KB333K3	100	0.033	0.235	0.750	0.688
CQ10A-KB823K3	100	0.082	0.312	0.875	0.812
CQ10A-KB104K3	100	0.10	0.312	0.875	0.812
CQ10A-KB124K3	100	0.12	0.400	0.875	0.812
CQ10A-KB154K3	100	0.15	0.400	0.875	0.812
CQ10A-KB184K3	100	0.18	0.400	1.125	1.062
CQ10A-KB224K3	100	0.22	0.400	1.125	1.062
CQ10A-KB274K3	100	0.27	0.400	1.375	1.312
CQ10A-KB394K3	100	0.39	0.562	1.125	1.062
CQ10A-KB474K3	100	0.47	0.562	1.125	1.062
CQ10A-KB684K3	100	0.68	0.562	1.375	1.312
CQ10A-KB105K3	100	1.0	0.562	1.875	1.812
CQ10A-KC123K3	200	0.012	0.235	0.750	0.688
CQ10A-KC153K3	200	0.015	0.235	0.750	0.688
CQ10A-KC223K3	200	0.022	0.235	0.750	0.688
CQ10A-KC473K3	200	0.047	0.312	0.875	0.812
CQ10A-KC563K3	200	0.056	0.312	0.875	0.812
CQ10A-KC683K3	200	0.068	0.312	0.875	0.812
CQ10A-KC823K3	200	0.082	0.400	0.875	0.812
CQ10A-KC104K3	200	0.10	0.400	0.875	0.812
CQ10A-KC124K3	200	0.12	0.400	1.125	1.062
CQ10A-KC154K3	200	0.15	0.400	1.125	1.062
CQ10A-KC184K3	200	0.18	0.400	1.375	1.312
CQ10A-KC224K3	200	0.22	0.400	1.375	1.312
CQ10A-KC274K3	200	0.27	0.562	1.125	1.062
CQ10A-KC334K3	200	0.33	0.562	1.125	1.062
CQ10A-KC474K3	200	0.47	0.562	1.375	1.312
CQ10A-KC564K3	200	0.56	0.562	1.875	1.812
CQ10A-KC684K3	200	0.68	0.562	1.875	1.812
CQ10A-KC824K3	200	0.82	0.670	1.875	1.812
CQ10A-KC105K3	200	1.0	0.670	1.875	1.812
CQ10A-KE822K3	400	0.0082	0.235	0.750	0.688
CQ10A-KE103K3	400	0.010	0.235	0.750	0.688
CQ10A-KE333K3	400	0.033	0.312	0.875	0.812
CQ10A-KE393K3	400	0.039	0.400	0.875	0.812
CQ10A-KE473K3	400	0.047	0.400	0.875	0.812
CQ10A-KE563K3	400	0.056	0.400	1.125	1.062
CQ10A-KE683K3	400	0.068	0.400	1.125	1.062
CQ10A-KE104K3	400	0.10	0.400	1.375	1.312
CQ10A-KE154K3	400	0.15	0.562	1.125	1.062
CQ10A-KE184K3	400	0.18	0.562	1.375	1.312
CQ10A-KE224K3	400	0.22	0.562	1.375	1.312
CQ10A-KE274K3	400	0.27	0.562	1.875	1.812
CQ10A-KE334K3	400	0.33	0.562	1.875	1.812
CQ10A-KE394K3	400	0.39	0.670	1.875	1.812
CQ10A-KE474K3	400	0.47	0.670	1.875	1.812
CQ10A-KE564K3	400	0.56	0.750	2.125	2.062
CQ10A-KE684K3	400	0.68	0.750	2.125	2.062
CQ10A-KE824K3	400	0.82	1.000	1.875	1.812
CQ10A-KE105K3	400	1.0	1.000	1.875	1.812
CQ10A-KF102K3	600	0.001	0.235	0.750	0.688
CQ10A-KF152K3	600	0.0015	0.235	0.750	0.688
CQ10A-KF222K3	600	0.0022	0.235	0.750	0.688
CQ10A-KF332K3	600	0.0033	0.235	0.750	0.688
CQ10A-KF472K3	600	0.0047	0.235	0.750	0.688

1/ See footnote at end of table.

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TABLE II. Style CQ10 capacitors. - Continued

Part number 1/	DC voltage rating	Capacitance	Case dimensions (in inches)		
			D +.010 -.020	L ±.015	
				Circuit 1	Circuit 2
	Volts	µF			
CQ10A-KF562K3	600	0.0056	0.235	0.750	0.688
CQ10A-KF682K3	600	0.0068	0.235	0.750	0.688
CQ10A-KF103K3	600	0.01	0.312	0.875	0.812
CQ10A-KF153K3	600	0.015	0.312	0.875	0.812
CQ10A-KF223K3	600	0.022	0.312	0.875	0.812
CQ10A-KF333K3	600	0.033	0.400	0.875	0.812
CQ10A-KF473K3	600	0.047	0.400	1.125	1.062
CQ10A-KF683K3	600	0.068	0.400	1.375	1.312
CQ10A-KF104K3	600	0.10	0.562	1.125	1.062
CQ10A-KF124K3	600	0.12	0.562	1.375	1.312
CQ10A-KF154K3	600	0.15	0.562	1.875	1.312
CQ10A-KF184K3	600	0.18	0.562	1.875	1.812
CQ10A-KF224K3	600	0.22	0.562	1.875	1.812
CQ10A-KF274K3	600	0.27	0.670	1.875	1.812
CQ10A-KF334K3	600	0.33	0.670	1.875	1.812
CQ10A-KF394K3	600	0.39	0.750	2.125	2.062
CQ10A-KF474K3	600	0.47	0.750	2.125	2.062
CQ10A-KF684K3	600	0.68	1.000	1.875	1.812
CQ10A-KG153K3	1,000	0.015	0.400	0.875	0.812
CQ10A-KG223K3	1,000	0.022	0.400	1.125	1.062
CQ10A-KG273K3	1,000	0.027	0.400	1.375	1.312
CQ10A-KG333K3	1,000	0.033	0.400	1.375	1.312
CQ10A-KG473K3	1,000	0.047	0.562	1.125	1.062
CQ10A-KG563K3	1,000	0.056	0.562	1.375	1.312
CQ10A-KG823K3	1,000	0.082	0.562	1.625	1.562
CQ10A-KG104K3	1,000	0.1	0.562	1.625	1.562
CQ10A-KG154K3	1,000	0.15	0.670	1.625	1.562
CQ10A-KG224K3	1,000	0.22	0.750	2.125	2.062
CQ10A-KG274K3	1,000	0.27	0.750	2.625	2.562
CQ10A-KG334K3	1,000	0.33	0.750	2.625	2.562
CQ10A-KG474K3	1,000	0.47	1.000	2.125	2.062

1/ Complete part number shall include an additional symbol to indicate circuit.

TABLE III. Millimeter equivalents of decimal inches.

Inches	mm	Inches	mm	Inches	mm
0.010	0.25	0.750	19.05	1.625	41.28
0.015	0.38	0.812	20.62	1.812	46.02
0.020	0.51	0.875	22.23	1.875	47.63
0.235	5.97	1.000	25.40	2.062	52.37
0.312	7.92	1.062	26.97	2.125	53.98
0.400	10.16	1.125	28.58	2.562	65.07
0.562	14.27	1.312	33.32	2.625	66.68
0.670	17.02	1.375	34.93		
0.688	17.48	1.562	39.67		

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Stability at low and high temperature:
Low temperature:
Test temperature - $-65^{\circ}{}^{+0^{\circ}}_{-5^{\circ}}$C.

Capacitance change (max) - -10 percent.
High temperature:
Test temperature - $+125^{\circ}{}^{+5^{\circ}}_{-0^{\circ}}$C.

Capacitance change (max) - +10 percent.

Life:
Capacitance change (max) - ±5 percent.

Resistance to soldering heat: In accordance with MIL-C-19978.

NOTE: CP10 capacitors formerly covered in MIL-C-25/1K are substitutable for capacitors covered in this specification sheet as follows:

FROM MIL-C-25/1K (CP10)	TO MIL-C-19978/20 (CQ10)	FROM MIL-C-25/1K (CP10)	TO MIL-C-19978/20 (CQ10)
CP10A-KB333K-	CQ10A-KB333K3	CP10A-KE334K-	CQ10A-KE334K3
CP10A-KB823K-	CQ10A-KB823K3	CP10A-KE394K-	CQ10A-KE394K3
CP10A-KB104K-	CQ10A-KB104K3	CP10A-KE474K-	CQ10A-KE474K3
CP10A-KB124K-	CQ10A-KB124K3	CP10A-KE564K-	CQ10A-KE564K3
CP10A-KB154K-	CQ10A-KB154K3	CP10A-KE684K-	CQ10A-KE684K3
CP10A-KB184K-	CQ10A-KB184K3	CP10A-KE824K-	CQ10A-KE824K3
CP10A-KB224K-	CQ10A-KB224K3	CP10A-KE105K-	CQ10A-KE105K3
CP10A-KB274K-	CQ10A-KB274K3	CP10A-KF102K-	CQ10A-KF102K3
CP10A-KB394K-	CQ10A-KB394K3	CP10A-KF152K-	CQ10A-KF152K3
CP10A-KB474K-	CQ10A-KB474K3	CP10A-KF222K-	CQ10A-KF222K3
CP10A-KB684K-	CQ10A-KB684K3	CP10A-KF332K-	CQ10A-KF332K3
CP10A-KB105K-	CQ10A-KB105K3	CP10A-KF472K-	CQ10A-KF472K3
CP10A-KC123K-	CQ10A-KC123K3	CP10A-KF562K-	CQ10A-KF562K3
CP10A-KC153K-	CQ10A-KC153K3	CP10A-KF682K-	CQ10A-KF682K3
CP10A-KC223K-	CQ10A-KC223K3	CP10A-KF103K-	CQ10A-KF103K3
CP10A-KC473K-	CQ10A-KC473K3	CP10A-KF153K-	CQ10A-KF153K3
CP10A-KC563K-	CQ10A-KC563K3	CP10A-KF223K-	CQ10A-KF223K3
CP10A-KC683K-	CQ10A-KC683K3	CP10A-KF333K-	CQ10A-KF333K3
CP10A-KC823K-	CQ10A-KC823K3	CP10A-KF473K-	CQ10A-KF473K3
CP10A-KC104K-	CQ10A-KC104K3	CP10A-KF683K-	CQ10A-KF683K3
CP10A-KC124K-	CQ10A-KC124K3	CP10A-KF104K-	CQ10A-KF104K3
CP10A-KC154K-	CQ10A-KC154K3	CP10A-KF124K-	CQ10A-KF124K3
CP10A-KC184K-	CQ10A-KC184K3	CP10A-KF154K-	CQ10A-KF154K3
CP10A-KC224K-	CQ10A-KC224K3	CP10A-KF184K-	CQ10A-KF184K3
CP10A-KC274K-	CQ10A-KC274K3	CP10A-KF224K-	CQ10A-KF224K3
CP10A-KC334K-	CQ10A-KC334K3	CP10A-KF274K-	CQ10A-KF274K3
CP10A-KC474K-	CQ10A-KC474K3	CP10A-KF334K-	CQ10A-KF334K3
CP10A-KC564K-	CQ10A-KC564K3	CP10A-KF394K-	CQ10A-KF394K3
CP10A-KC684K-	CQ10A-KC684K3	CP10A-KF474K-	CQ10A-KF474K3
CP10A-KC824K-	CQ10A-KC824K3	CP10A-KF684K-	CQ10A-KF684K3
CP10A-KC105K-	CQ10A-KC105K3	CP10A-KG153K-	CQ10A-KG153K3
CP10A-KE822K-	CQ10A-KE822K3	CP10A-KG223K-	CQ10A-KG223K3
CP10A-KE103K-	CQ10A-KE103K3	CP10A-KG273K-	CQ10A-KG273K3
CP10A-KE333K-	CQ10A-KE333K3	CP10A-KG333K-	CQ10A-KG333K3
CP10A-KE393K-	CQ10A-KE393K3	CP10A-KG473K-	CQ10A-KG473K3
CP10A-KE473K-	CQ10A-KE473K3	CP10A-KG563K-	CQ10A-KG563K3
CP10A-KE563K-	CQ10A-KE563K3	CP10A-KG823K-	CQ10A-KG823K3
CP10A-KE683K-	CQ10A-KE683K3	CP10A-KG104K-	CQ10A-KG104K3
CP10A-KE104K-	CQ10A-KE104K3	CP10A-KG154K-	CQ10A-KG154K3
CP10A-KE154K-	CQ10A-KE154K3	CP10A-KG224K-	CQ10A-KG224K3
CP10A-KE184K-	CQ10A-KE184K3	CP10A-KG274K-	CQ10A-KG274K3
CP10A-KE224K-	CQ10A-KE224K3	CP10A-KG334K-	CQ10A-KG334K3
CP10A-KE274K-	CQ10A-KE274K3	CP10A-KG474K-	CQ10A-KG474K3

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